

Project Initialization and Planning Phase

Date	5-7-2026
Team ID	
Project Title	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution)

The proposed system is an AI-powered learning platform that helps students understand concepts, answer academic questions, generate quizzes, create short summaries, and follow personalized learning paths. EduGenie improves the learning experience by providing fast, reliable, and interactive educational support.

Project Overview	
Objective	Build an intelligent learning assistant that delivers accurate answers, easy-to-understand explanations, practice quizzes, concise summaries, and customized learning paths to support effective learning.
Scope	The project develops a web-based educational assistant using FastAPI, HTML, CSS, Jinja2, SQLite, and Google Gemini AI. It enables students to learn efficiently through interactive AI-powered features.

Problem Statement	
Description	Many students struggle to find reliable study material, understand difficult concepts, prepare for exams, and organize their learning using multiple online resources.
Impact	The application reduces study time, improves concept clarity, increases learning efficiency, and helps students prepare with greater confidence.

Proposed Solution	
Approach	Develop a FastAPI-based web application integrated with Google Gemini AI to provide intelligent educational support through a simple and responsive interface.
Key Features	• AI-powered Question Answering • Beginner-Friendly Concept Explanations • Interactive Quiz Generation • Smart Text Summarization • Personalized Learning Paths • Responsive Web Interface

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	Processor requirement	Intel Core i5 / AMD Ryzen 5 or above
Memory	Minimum memory required	8 GB RAM or higher
Storage	Storage required for application files	256 GB SSD
Software		
Framework	Web application framework	FastAPI
Libraries	Python packages	google-generativeai, uvicorn, jinja2, python-dotenv, python-multipart, pytest
Development Environment	Code editor	Visual Studio Code
Data		
Database	Local database	SQLite
AI Model	AI language model	Google Gemini 2.5 Flash